

Question B2

This question carries 20 marks in total.

(a) Given the potential field

$$\phi = x^2 z^3 + e^{xy}$$

and the point P with coordinates $(1, 0, 1)$, find

(i) [4 MARKS] $\nabla\phi$ at P ;

(ii) [4 MARKS] the rate of change of ϕ in the direction $\hat{v}$ where $\mathbf{v} = 2\mathbf{i} + \mathbf{j} + 2\mathbf{k}$.

(b) (i) [4 MARKS] Use Cartesian coordinates to verify that an arbitrary (differentiable) scalar field ϕ satisfies the vector identity

$$\nabla \times \nabla\phi = \mathbf{0}.$$

(ii) [4 MARKS] The vector field $\mathbf{F}$ is given by

$$\mathbf{F} = 2xy\mathbf{i} + x^2\mathbf{j} + 2z\mathbf{k}.$$

Calculate $\nabla \times \mathbf{F}$. What can you deduce about the vector field $\mathbf{F}$?

(iii) [4 MARKS] Evaluate the line integral of the vector field $\mathbf{F}$, given above, between the points $(1, 1, 2)$ and $(2, 1, 0)$. Why does the answer not depend on the path taken?